

Deep Learning and Artificial Neural Network (ANN)

1. What is Deep Learning?

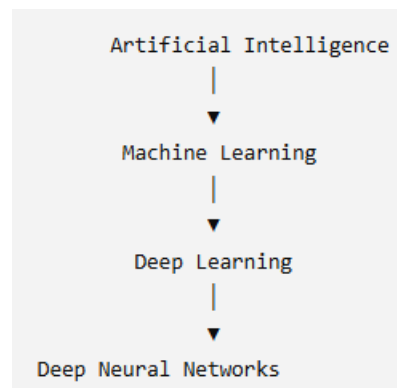
Deep Learning is a branch of Machine Learning (ML) that uses artificial neural networks with multiple layers to learn complex patterns from large amounts of data.

The term "**deep**" refers to the presence of multiple layers between the input and output.

Simple Definition

Deep Learning is a machine learning technique in which multi-layer neural networks automatically learn useful features and patterns from data to make predictions or decisions.

Machine Learning vs. Deep Learning



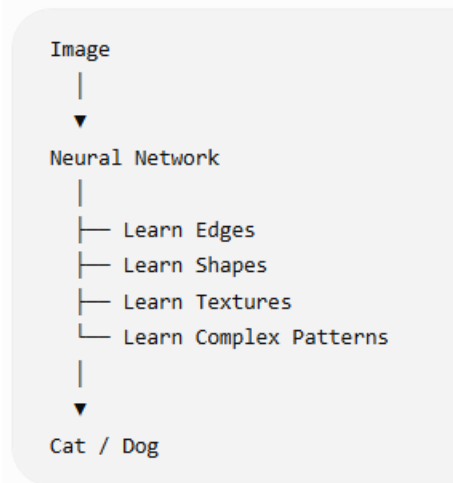
Example: Image Classification

Suppose we want to identify whether an image contains a **cat** or **dog**.

Traditional Machine Learning



Deep Learning

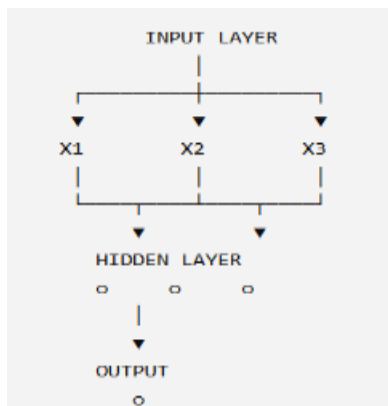


3. What is an Artificial Neural Network?

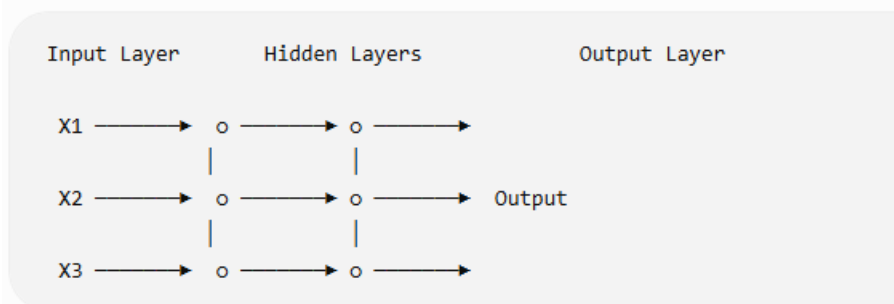
An Artificial Neural Network (ANN) is a computational model inspired by the way biological neurons process information.

An ANN consists of interconnected artificial neurons (nodes) arranged into layers.

The basic structure is:



A more complete neural network can contain several hidden layers:



4. Components of an Artificial Neural Network

An ANN mainly contains:

1. Input layer
2. Hidden layer(s)
3. Output layer
4. Weights
5. Bias
6. Activation function

4.1 Input Layer

The input layer receives the input features.

For example, suppose we want to predict whether a student will pass based on:

- Study hours
- Attendance
- Assignment score

Then:

$$X_1 = \text{Study Hours}$$

$$X_2 = \text{Attendance}$$

$$X_3 = \text{Assignment Score}$$

The input layer contains three input nodes.

4.2 Hidden Layer

The hidden layers process the input information.

A neural network can have:

- One hidden layer
- Two hidden layers
- Many hidden layers

When there are many hidden layers, the network is generally called a Deep Neural Network (DNN).

4.3 Output Layer

The output layer produces the final prediction.

For binary classification:

Output = 0 → Fail

Output = 1 → Pass

For multiclass classification:

Output 1 → Cat

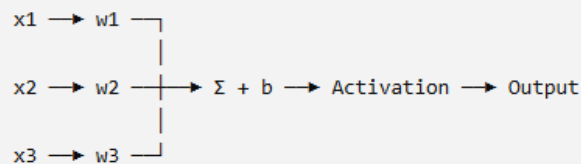
Output 2 → Dog

Output 3 → Horse

5. Artificial Neuron

The basic building block of an ANN is an **artificial neuron**.

A neuron receives inputs, multiplies them by weights, adds a bias, and passes the result through an activation function.



Mathematically:

$$z = w_1x_1 + w_2x_2 + w_3x_3 + b$$

Then:

$$y = f(z)$$

where:

- x_1, x_2, x_3 = inputs
- w_1, w_2, w_3 = weights
- b = bias
- f = activation function
- y = output

6. Numerical Example of an Artificial Neuron

Suppose:

$$x_1 = 2$$

$$x_2 = 3$$

Weights:

$$w_1 = 0.5$$

$$w_2 = 0.8$$

Bias:

$$b = -1$$

First calculate the weighted sum:

$$z = w_1x_1 + w_2x_2 + b$$

$$z = (0.5)(2) + (0.8)(3) - 1$$

$$z = 1 + 2.4 - 1$$

$$z = 2.4$$

If we use the **ReLU** activation function:

$$\text{ReLU}(z) = \max(0, z)$$

Therefore:

$$\text{ReLU}(2.4) = 2.4$$

So the neuron produces:

$$\boxed{\text{Output} = 2.4}$$

7. What is a Weight?

A **weight** determines how strongly an input influences a neuron.

For example:

$$z = w_1x_1 + w_2x_2 + b$$

If w_1 is large, x_1 has a greater influence on the output.

The weights are **learned automatically during training**.

8. What is Bias?

A **bias** allows a neuron to shift its activation function.

The formula is:

$$z = w_1x_1 + w_2x_2 + b$$

Without bias:

$$z = w_1x_1 + w_2x_2$$

With bias:

$$z = w_1x_1 + w_2x_2 + b$$

Bias gives the neural network additional flexibility when learning patterns.

9. Activation Function

An activation function determines the output of a neuron after calculating the weighted sum.

Common activation functions include:

ReLU

$$ReLU(x) = \max(0, x)$$

Example:

$$ReLU(-2) = 0$$

$$ReLU(3) = 3$$

Sigmoid

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

Its output lies between **0 and 1**, so it is commonly used for binary classification output layers.

Tanh

$$\tanh(x)$$

Its output lies between **-1 and +1**.

Softmax

Softmax is commonly used in the output layer for **multiclass classification**.

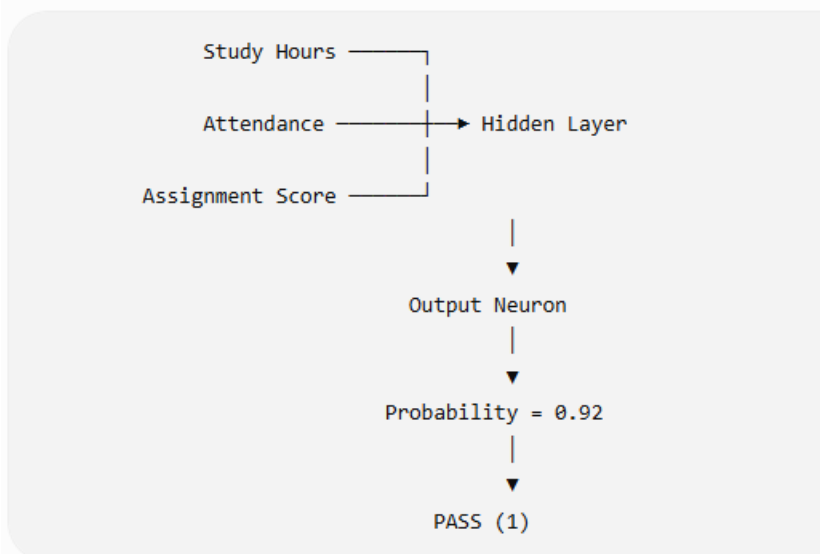
10. ANN Example: Predicting Student Result

Suppose we want to predict whether a student will pass.

Input

Feature	Value
Study Hours	6
Attendance	85%
Assignment Score	80

These values are supplied to the input layer.



If the output probability is:

$$P(Pass) = 0.92$$

and the classification threshold is 0.5:

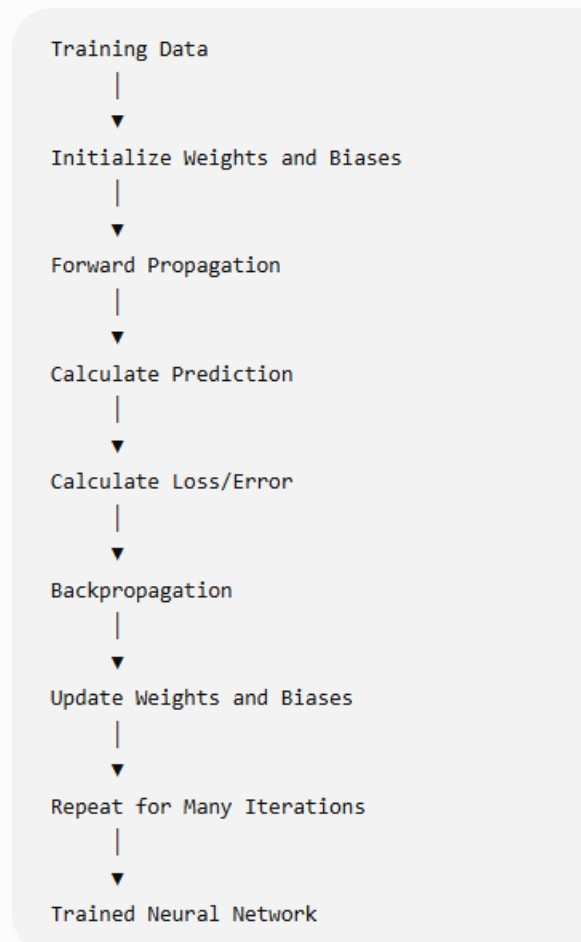
$$0.92 > 0.5$$

Therefore:

$$\boxed{Prediction = Pass}$$

11. How Does an ANN Learn?

The learning process can be summarized as:



Step 1: Forward Propagation

The input moves from the input layer toward the output layer.

Input → Hidden Layer → Output

The network calculates a prediction.

Step 2: Calculate Loss

The prediction is compared with the actual answer.

For example:

Actual:

$$y = 1$$

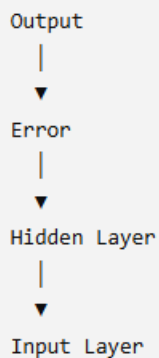
Predicted:

$$\hat{y} = 0.8$$

A loss function measures how far the prediction is from the actual value.

Step 3: Backpropagation

The error is propagated backward through the network.



The algorithm determines how the weights contributed to the error.

Step 4: Update Weights

An optimization algorithm such as **Gradient Descent** updates the weights and biases.

A simplified update rule is:

$$w_{new} = w_{old} - \eta \frac{\partial L}{\partial w}$$

where:

- w = weight
- L = loss
- η = learning rate

This process is repeated until the model learns a useful representation of the training data.

13. Applications of Deep Learning and ANN

Image Recognition

Image → CNN → Features → Classification

Used for:

- Object recognition
- Face detection
- Medical image analysis

Speech Recognition

Audio → Neural Network → Text

Natural Language Processing

Used in:

- Machine translation
- Text classification
- Question answering
- Speech and language applications

Healthcare

Used for:

- Medical image classification
- Risk prediction
- Pattern detection in medical data

Finance

Used for:

- Fraud detection
- Credit risk analysis
- Financial prediction

14. Advantages of Neural Networks

- Can learn complex nonlinear relationships.
- Can automatically learn useful features, especially in deep networks.
- Works with images, text, audio, and numerical data.
- Can achieve high predictive performance with sufficient data and appropriate training.
- Can be used for classification, regression, and many other tasks.

15. Limitations

- Requires appropriate training data.
- Deep networks can require significant computational resources.
- Training can take considerable time.
- The learned model can be difficult to interpret.
- Hyperparameters such as learning rate, number of layers, and number of neurons need to be selected appropriately.

Final Summary

Deep Learning is a subset of Machine Learning that uses neural networks with multiple layers to learn complex patterns automatically.

An **Artificial Neural Network (ANN)** consists of:

Input Layer → Hidden Layer(s) → Output Layer

Each neuron performs:

$$z = \sum wx + b$$

followed by an activation function:

$$Output = f(z)$$

During training, the ANN uses **forward propagation**, **loss calculation**, **backpropagation**, and **weight/bias updates** to learn from the training data.

